

Netflix Content Analytics

Exploratory Data Analysis of **8,807 Netflix title records** — catalog composition, geography, ratings, genres, TV-show seasons, and data quality.

BUSINESS INTELLIGENCE

DATA SCIENCE EDA

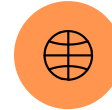


Business Objectives



Catalog Mix

Analyze Movie vs. TV Show composition.



Geography

Identify countries leading in content volume.



Ratings & Genres

Understand dominant ratings and genres.



TV Seasons

Explore TV-show season distribution patterns.



Release Trends

Examine release-year patterns across the catalog.



BI & ML Roadmap

Translate EDA findings into a BI dashboard and future ML phase.

Dataset at a Glance

8,807

Total Titles

Records analyzed in the dataset

6,131

Movies

69.6% of catalog — movie-dominated

2,666

TV Shows

30.3% — smaller but significant share

3,642

US Titles

United States — dominant content origin

India follows with **972 titles**, the second-largest country group. The dataset spans **1925–2021**, with **2018** as the peak release year at **1,146 titles**.

EDA Workflow

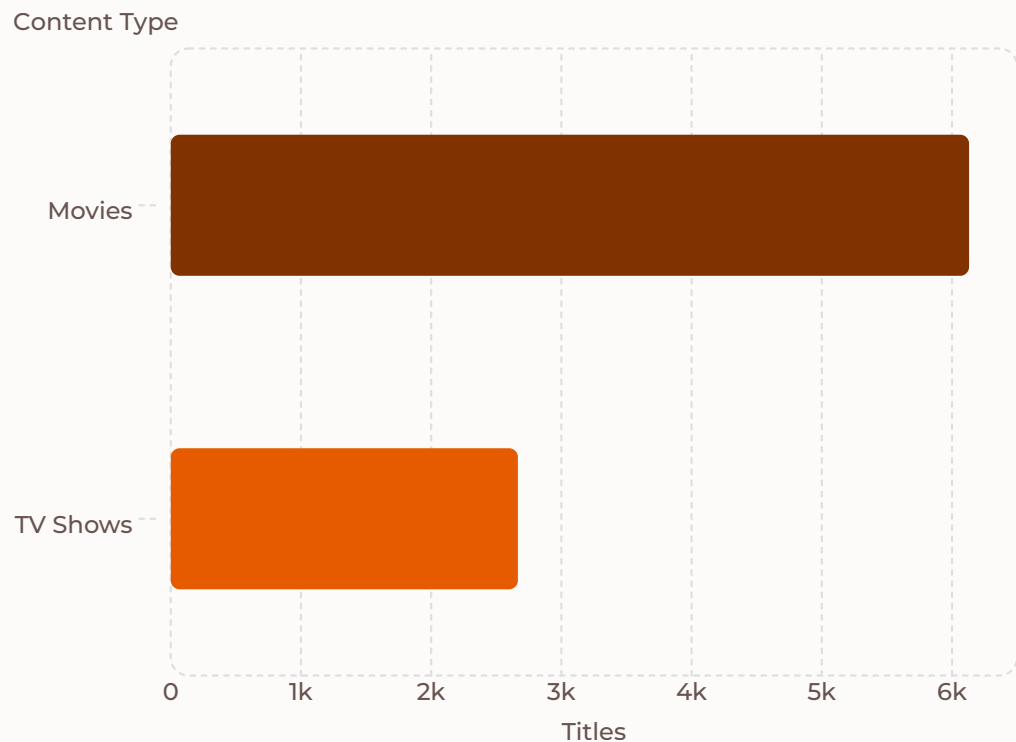


Tools & Technologies

The workflow follows **Load → Clean → Transform → Visualize → Interpret**, using pandas, numpy, matplotlib, and seaborn.

- **pandas** — data loading and manipulation
- **numpy** — numerical operations
- **matplotlib & seaborn** — visualization
- **warnings** — suppress non-critical alerts

Catalog Composition



Movie-Dominated Catalog

Movies account for nearly **70%** of all records, with 6,131 titles versus 2,666 TV shows — a 2.3x ratio that shapes every downstream analysis.

- This imbalance should be considered when comparing content-type metrics or building recommendation models.



Geographic Distribution

 United States — 3,642

Dominant content origin, representing the largest share of the catalog.

 India — 972

Second-largest country group, reflecting strong regional production.

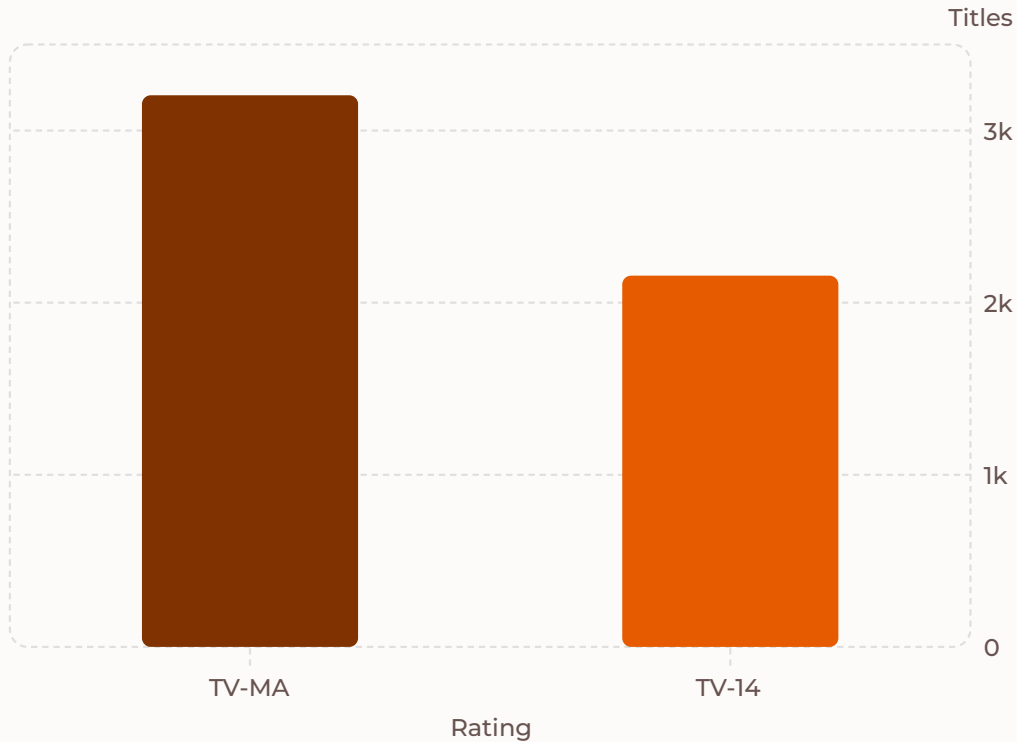
 United Kingdom — 418

Third-place contributor with notable volume.

 Japan — 244 ·  South Korea — 199

Notable Asian markets with meaningful catalog presence.

Ratings & Genres



Mature-Audience Presence

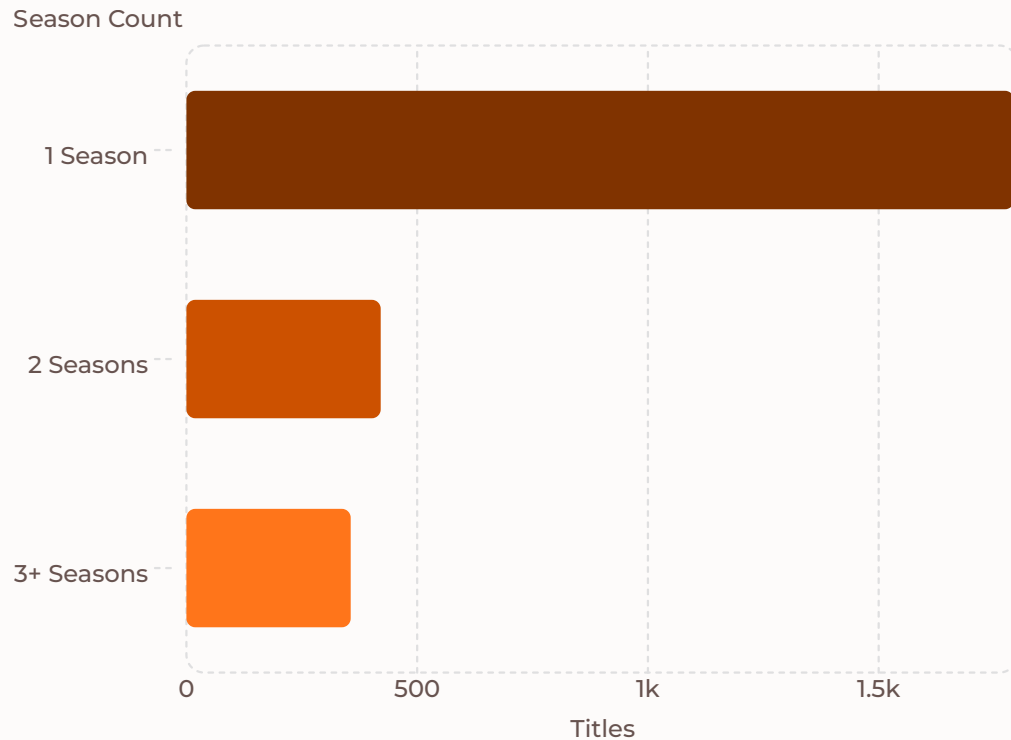
TV-MA (3,205) and **TV-14** (2,157) are the most frequent ratings, indicating a strong mature-audience focus in the catalog.

Leading Genres

- International Movies
- Dramas
- Comedies
- International TV Shows
- Documentaries

TV Seasons & Release Years

TV Show Seasons



Release Year Trends

The dataset spans **1925–2021**. **2018** is the peak year with **1,146 titles**, followed by a visible decline from 2020 onward.

- Single-season shows dominate TV content at 1,793 titles — over 4x the volume of multi-season shows.

Data Quality & Limitations

Field	Missing Values	Implication
Director	2,634 missing	Mode imputation distorts frequency-based analysis
Cast	825 missing	Actor/cast analysis may be incomplete
Country	831 missing	Geographic analysis has gaps
date_added	10 missing	Datetime features require validation
Rating	4 missing	Small gap in audience segmentation
Duration	3 missing	Minor gap for duration analysis

Key Findings from Initial EDA



Content Type Bias

Movies dominate the catalog at nearly 70% (6,131 titles), significantly outweighing TV shows (30% or 2,666 titles).



Geographic Concentration

The **United States leads** with 3,642 titles, followed by India (972) and the United Kingdom (418), highlighting key content production hubs.



Mature Audience Focus

TV-MA (3,205 titles) and **TV-14** (2,157 titles) are the most frequent content ratings, indicating a strong emphasis on mature programming.



Top Genres

Leading genres include **International Movies, Dramas, Comedies, International TV Shows, and Documentaries.**



Single-Season TV Shows

Most TV content consists of **single-season shows** (1,793 titles), significantly outnumbering multi-season series.



Recent Production Decline

Content releases peaked in **2018** (1,146 titles), showing a visible **decline from 2020** onwards.